



DEPARTMENT OF COMPUTER SCIENCE

COMP438 - Distributed Computing

Course Project - phase 1

Dr. Mohammad Helal

Project Summary:

Peer-to-Peer chat application using a Directory server to help clients exchange network credentials (IP and Port numbers).

Project Details:

Your project should be done using Java and it should be organized as follows:

1. Client process:

- Client programs allows user to chat with other clients. Client process establishes TCP connection with Dictionary process (will be explained next) upon activation, it sends a periodic "I am alive" message to the Directory every one second, and Directory saves the IP and Port numbers for that client when receiving "I am alive" message.
- Client process receives a list of all active clients periodically (names if IDs only) from the Directory process, then the user should be able to select a client to chat with, when selected, it asks Directory to send IP and Port numbers for the destination client.
- Client process needs to be multithreaded so it can exchange messages with other clients asynchronously.

2. Dictionary process :

- it links between the clients
- it keeps track of which clients are available.
- it sends the list of available clients periodically (only the names)
- when a client requests a connection with another client, Dictionary sends IP and Port

for destination.

What to Submit?

1. Only .java files for Client, and Dictionary files
2. Report: your report should explain:
 - the design of your project
 - how many sockets needed, how did you manage concurrency
 - explain how client established connection with other clients
 - explain the application layer header your project proposes